

L10: Lesson Summary



This lesson focused on information diffusion processes on networks.

We started with some common empirical findings about the size and depth of the cascades that these diffusion processes create .

Then, we reviewed some common mathematical models, such as the Independent Cascade model, that are often used to study such network diffusion processes.

Then, we examined an important optimization problem in the context of network diffusion: if we can start a cascade from say k nodes, which nodes should we select to maximize the size of the cascade?

We also considered the effect of community structure on the extent of the diffusion process on a network.

Finally, we presented a case study of a network diffusion process on brain networks, in the context of multi-sensory integration.